

Shib Kumar Saraf

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Kolkata, India • Open to Relocation

AI Engineer with 3+ years shipping production GenAI systems — including an LLM inference platform serving **10,000+ monthly queries** on India's PARAM Siddhi supercomputer, and a PHI/PII-aware document-intelligence platform for enterprise data.

Skills

- **GenAI & LLMs:** RAG Architectures, LangChain, LangGraph, Prompt Engineering, Fine-Tuning, Vector Search (Pinecone, ChromaDB), Model Serving (vLLM, Ollama), OpenAI / Gemini / Hugging Face APIs.
- **ML, NLP & Computer Vision:** Predictive Modeling, NLP (NER, Classification), Computer Vision, PII/PHI Detection & Redaction, Time-Series Forecasting.
- **Data Engineering:** Python, SQL, PySpark, ETL Pipelines, Data Cleaning & EDA, BigQuery, Databricks.
- **Platforms & Infra:** FastAPI, REST APIs, Docker, AWS, GCP, HPC (Slurm/PBS, GPU Acceleration), Linux.

Experience

- **EXL Service (India) Private Limited** Oct 2025 – Present
Gen AI Engineer
 - Engineered a GenAI document-extraction pipeline that parses unstructured Aetna records and extracts structured cost-sharing fields (copay, deductible, coinsurance) into a unified data layer, powering downstream lead-generation and analytics.
 - Collaborated on a document-intelligence POC, Digital Binder, that auto-classifies uploaded files across 6+ document types (CVs, licenses, IDs, etc.), extracts full content, and identifies PHI/PII fields for automated labeling.
 - Designed an RBAC-driven masking layer that dynamically redacts PHI/PII based on requester role, then layered a retrieval-based chatbot on top so users can query document contents in natural language while the same masking rules are enforced at query time.
 - Built statistical churn-prediction models across demographic and geographic factors for CVS Health (Aetna) and presented findings to plan sponsors and cross-functional stakeholders, contributing to a retention increase from 49% to 60%.
- **Centre for Development of Advanced Computing (C-DAC), Chennai** Jul 2024 – Oct 2025
Project Associate (AI Engineer)
 - Shipped two production RAG chatbot platforms (ChatDB, FileQAExpert) — covering document ingestion, retrieval, and GPU-optimized vLLM inference (320 ms latency, 2.8s response time), deployed on India's **PARAM Siddhi** supercomputing cluster, serving **10,000+ monthly queries** across government departments.
 - Contributed to the development of Secure Exam Analytics, a real-time analytics platform combining computer-vision based anomaly detection, behavioral profiling, and collusion detection for high-stakes government examinations, cutting manual review effort by **60%** and improving monitoring efficiency by **40%**.
 - Engineered secure, end-to-end financial-workflow automation for claims processing and compliance reporting under the **QDAS** project, ensuring data integrity across the pipeline.
- **Ministry of Electronics and Information Technology (Govt. of India)** Jan 2024 – Jun 2024
Big Data & Data Analytics Intern
 - Designed and implemented **Social Sentinel**, a real-time NLP-powered system for monitoring, classifying, and flagging offensive content on Telegram, backed by scalable data-ingestion pipelines and analytics dashboards.
 - Participated in applied NLP and ML research on clinical risk prediction under **Health Analytics Project**, leveraging transformer models and topic modeling techniques on real-world Chinese hospital data.
- **Solvitude Solutions Private Limited** May 2021 – Nov 2021
Subject Matter Expert & Quality Assurance Analyst
 - Automated reporting workflows and created structured question banks and analytical reports to streamline product evaluation, training, and QA processes, while testing and documenting hardware and software systems with cross-functional teams.

Research Publications

- **Multifaceted Feature Extraction and Machine Learning for Mortality Risk Prediction in Pediatric ICUs**
DOI: 10.1109/ICSTSDG61998.2024.11026232 *IEEE ICSTSDG 2024, Chennai, India*
An ML-driven clinical risk prediction framework using advanced feature engineering to improve mortality assessment in pediatric intensive care units.

- **Hybrid QLSTM Framework for Time-Series Forecasting in Dynamic Financial Markets**
DOI: 10.1109/ICIMIA67127.2025.11200988 *IEEE ICIMIA 2025, Tirupur, India*
A hybrid Quantum-inspired LSTM architecture to improve forecasting accuracy under highly volatile financial market conditions.
- **Real-Time Vigilance: Advanced Video Analysis for Violence Detection in Public Spaces**
DOI: 10.1109/ICRITO66076.2025.11241501 *IEEE ICRITO 2025, Noida NCR, India*
A real-time computer vision pipeline leveraging deep learning models for automated violence detection to support public safety monitoring systems.

Personal Project

- **QueryCortex AI** — Multi-Tenant Agentic RAG Platform   Oct 2025 – Dec 2025
 - Implemented a multi-tenant agentic RAG architecture combining NL2SQL and vector retrieval, enabling natural-language queries across structured databases and unstructured documents with explainable, source-cited outputs.
 - Secured the platform with JWT-based authentication, document versioning and FastAPI-driven API orchestration, enabling safe enterprise data access across tenants.
- **CodeGuard AI** — AI-Powered Secure Code Review Platform  Jan 2025 – Mar 2025
 - Engineered a full-stack **MERN** application integrating **Google Gemini** to analyze source code across **3+ programming languages**, detecting security vulnerabilities, code smells, performance bottlenecks, and maintainability issues with structured AI-generated recommendations.
- **MindMetrics** — Mental Health Insights & Behavioral Analytics  Feb 2024 – Apr 2024
 - Architected and built an end-to-end **LLM-powered NLP pipeline** combining BERT, Gemini, zero-shot classification, and topic modeling to analyze **20K+ social media posts** and surface behavioral patterns and emerging mental-health signals.
- **SereneSounds** — Personalized Music Curation Engine  Sep 2023 – Dec 2023
 - Developed a hybrid recommendation engine combining **collaborative filtering, ALS, graph-based similarity, lyrical embeddings, and audio feature representations** to generate personalized recommendations across a catalog of **8K+ songs** through an interactive Streamlit application.

Education

- **Pondicherry University** Nov 2022 – Jun 2024
Master of Computer Applications CGPA: 9.35/10
- **Indian Institute of Technology Madras** Aug 2021 – May 2023
Diploma in Data Science CGPA: 8.8/10
- **Panskura Banamali College, Vidyasagar University (Affiliated)** Jul 2019 – Aug 2022
B.Sc. in Physics (Honours) CGPA: 9.08/10

Certifications

- **Artificial Intelligence for Economics** — IIT Kharagpur (NPTEL, 2025)
- **Deep Learning for Computer Vision** — IIT Hyderabad (NPTEL, 2023)
- **SQL (Advanced) Certificate** — HackerRank (2023)

Achievements

- **Best Paper Award** — **IEEE ICRITO 2025**, for "Real-Time Vigilance: Advanced Video Analysis for Violence Detection in Public Spaces" (see Research Publications).
- **Best Project Award** — Awarded for the project in the *Modern Application Development* course at **IIT Madras**.
- **Top 3 Rank Holder** — Secured a position among the top 3 students in the MCA program at **Pondicherry University** with a CGPA of 9.35.